

ATIMANAS BARAL

Data Engineer · Quantitative Analyst · ML Engineer

Bengaluru, India | +91 98610 41381 | atimanasbaral@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Electrical & Electronics Engineering graduate (VSSUT, 2025) who builds systems that turn raw financial data into actionable signals — from live NSE options mispricing detection (**73% accuracy, XGBoost/LightGBM ensemble**) to a production Snowflake data warehouse processing **5M+ transactions daily**. Brings a rare combination of quant finance depth (VaR, Black-Scholes, backtesting) and engineering rigour (FastAPI microservices, INT8 model quantization, ETL pipelines). Currently sharpening Airflow and dbt for end-to-end pipeline orchestration. Targeting high-impact roles in data engineering, quant analytics, or ML engineering at fintech and HFT organizations in Bengaluru.

EDUCATION

B.Tech — Electrical & Electronics Engineering | VSSUT Burla Oct 2022 – Jun 2025
Relevant Coursework: Signals & Systems, Control Systems, Probability & Statistics, Linear Algebra, Numerical Methods, Microprocessors

TECHNICAL SKILLS

Languages	Python, SQL, R, C++ (learning)
ML / Analytics	Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, MLflow, ONNX, TensorRT, PSI Drift Detection, Ensemble Stacking, NLP, Time-Series Forecasting
Data Engineering	Snowflake, PostgreSQL, BigQuery, AWS, ETL Pipelines, Star Schema, Dimensional Modeling, SQL Optimization · Learning: Apache Airflow, dbt, PySpark
Quant Finance	Monte Carlo Simulation, VaR (95%/99% CI), Kupiec/Christoffersen Backtesting, Black-Scholes, Implied Volatility Surface, Greeks ($\Delta/\Gamma/V$), FIX Protocol, Financial APIs
Visualization	Power BI, Tableau, FastAPI Dashboards
DevOps / Tools	Docker, FastAPI, REST APIs, Git, Linux, JIRA, Agile

EXPERIENCE

Fraud Analyst — BlackBuck Sep 2025 – Mar 2026 | Bengaluru

- Processed and classified 170+ call recordings per day with fraud-indicator tagging, sustaining >95% labelling precision — directly reducing false-negative leakage in the fraud model pipeline.
- Engineered VLOOKUP-based cross-validation workflows across multi-source datasets, cutting reconciliation discrepancies by ~40% and improving data reliability feeding into risk-scoring models.
- Flagged 80+ high-risk user accounts per week through pattern-based anomaly detection, contributing to estimated fraud prevention and supporting compliance reporting workflows.
- Redesigned manual data validation into a structured batch-review process, reducing fraud-case review turnaround by ~30% and freeing analyst capacity for exception handling.
- Delivered structured fraud-signal reports to Risk, Ops, and Legal stakeholders, enabling data-driven policy decisions with zero critical data-quality escalations over tenure.

PROJECTS

Options Mispricing Hunter — ML over Black-Scholes Residuals 2025
Python · XGBoost · LightGBM · NSE Options API · FastAPI · Implied Volatility Surface · MLflow

- Accomplished 73% directional accuracy on 6-month out-of-sample NSE options data — exceeding the 51% baseline by 22 pp — by framing Black-Scholes residuals as an ML target and training an XGBoost/LightGBM ensemble over 25+ features including IV surface slope, Greeks (Δ , Γ , V), put-call parity deviation, and momentum signals.
- Reduced false-positive signal rate by 31% versus baseline by incorporating time-to-expiry bucketing and Sharpe ratio-weighted signal ranking, improving capital efficiency for live paper-trading trials.
- Tracked model performance across 3 retraining cycles using MLflow experiment logging, detecting IV surface regime shifts via PSI drift monitoring and triggering automated retraining pipelines.

Monte Carlo VaR Engine — NIFTY50 Portfolio Risk System2025

Python · NumPy · SciPy · Financial APIs · Kupiec/Christoffersen Backtesting

- Accomplished Basel II-compliant VaR (95%/99% CI) for a 20-asset NIFTY50 portfolio, running 10,000+ correlated Cholesky-decomposed simulations per asset in sub-second execution time across 3 years of historical data.
- Validated model under regulatory standards via Kupiec and Christoffersen backtests, achieving p-values above the 5% significance threshold across all 5 market stress regimes tested.
- Reduced manual risk analysis time by 80% through automated parameterized report generation with stress-testing (bull, bear, high-volatility, low-liquidity, crisis) via a six-panel interactive dashboard.

Snowflake E-Commerce Data Warehouse — Real-Time Analytics Platform2025

Snowflake · PostgreSQL · Star Schema · ETL Pipelines · SQL Optimization · Power BI

- Achieved 40% query performance improvement on a Snowflake warehouse ingesting 5M+ daily transactions by implementing star-schema dimensional modeling with partition pruning, clustering keys, and query restructuring across 20 reporting dashboards.
- Built automated batch and incremental ETL pipelines with data quality validation checkpoints, reducing pipeline failure rate to near-zero and ensuring SLA-compliant data freshness for downstream BI consumers.

CNN INT8 Quantization Pipeline — ResNet-50 / MobileNetV22024

PyTorch · ONNX · TensorRT · ImageNet · INT8 Post-Training Quantization

- Achieved 3.2× inference speedup and 60% model size reduction retaining 95.4% top-1 accuracy on ImageNet by designing an end-to-end INT8 pipeline (PyTorch → ONNX → TensorRT) for ResNet-50 and MobileNetV2.
- Reduced pipeline setup time from 6 hours to 45 minutes for new architectures by automating calibration across 4 precision modes (FP32, FP16, INT8, dynamic range) via a reusable CLI tool with YAML config.

LEADERSHIP & ACTIVITIES

Core Techno-Management Member — Idea and Innovation Cell, VSSUTNov 2022 – Sep 2024

- Designed and delivered 3 ML workshops for 80+ junior students; post-workshop feedback averaged 4.6/5 across all cohorts.
- Led cross-functional teams (5–8 members) to prototype 4 data-driven institutional solutions; 2 deployed within university infrastructure.
- Coordinated Samavesh Annual Fest 2023 — 500+ attendees, 3 technical tracks, 6 events including the flagship ML competition.

CERTIFICATIONS & SIMULATIONS

HackerRank SQL (Advanced)Jan 2026

Advanced querying, window functions, optimization · License: AA0F4E0EA6E7

Citi Finance Virtual Experience2024

Financial analysis, risk assessment, credit evaluation workflows

HSBC Banking Virtual Experience2024

Wholesale banking, transaction monitoring, compliance data analysis

Data Analysis with R Programming — Google / CourseraJan 2025

Statistical analysis, data wrangling, visualization in R

Introduction to Networking — Cisco Networking AcademyJun 2024

Network fundamentals, protocols, architecture